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PAPER_873 – U_g1=DPM Geophysical Geometry Summation

****CP4 Class:**** #457 `Ug1DPMGeophysicalGeometrySummationCalc`

****Session:**** 200C v5.61

****Source:**** `describe mass without using weight.txt` (3,094 lines)

Abstract

U_g1 is equivalent to the DPM; the total force is a summation over DPM variable forms re force), resonance (toroidal). Geometry functions G_k enable force diagramming.

Key Equations

1.
$$F_{Ug1} = \sum_k [k_k * (fUA1*fSCm1*REB1)*(fUA2*fSCm2*REB2)/r^2 * G_k]$$
2.
$$F_{SM_gravity} = k_{grav} * DPM1*DPM2/r^2 * \sin(\theta)$$
3.
$$F_{Ub} = k_{buoy} * (fUA'*fSCm)/r^2 * \sin(\theta)$$
4.
$$F_{resonance} = k_{res} * \nu_{THz} * fSCm * \cos(\phi)*f(\nu_{THz})$$
5.
$$F_{Ug1_total} = F_{grav} - F_{buoy} + F_{resonance}$$

UQFF Context

This calculator implements parameterized physics equations per the CondensedPhysics architecture (pure calculator, no hardcoded data). Inputs arrive from source2.cpp PRINCIPAL GUI via dataset dict. Outputs include primary_equations, available_equations, and simulation_set.

Source Thread

Grok 3 dialogue, Daniel Murphy, June 03-04 2025.
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Star-Magic UQFF Codebase – Session 200C

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.